SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

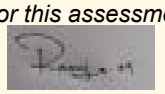
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:	CSA1716	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	Slot A / 2026	**Date:**	10 - 10 -2026

Code of Conduct

I Ramya M / 1921260148 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Read each question carefully; answer all four questions (100 in total). Off-Class Assessment Tool.
- Each question gives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement, component, iteration or step number), explain its root cause, correct it with justification and apply the corrected concept. An error merely carried forward counts once.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Error Analysis rubric: error identification, root cause analysis, correction and justification, application of concepts, and clarity and presentation.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	**Q1**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**

9. Uninformed Search Strategies	Error tracing of BFS, DFS and uniform-cost search	**Q3**
10. Search with Partial Information	Sensorless problems, belief states, percept update	**Q4**

Annexure A – Error Analysis Task

Question 1: Foundations of AI, Intelligent Agents and Rationality – Error Analysis of Concept Notes

Topic: Foundations and history of AI, agents and environments, rationality, nature of environments, structure of agents

A junior engineer prepared the revision notes below on the foundations of AI and intelligent agents. The notes contain errors; some statements may be correct. Analyse them statement by statement.

No.	Statement in the notes
S1	The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic. It is therefore the standard example of the "thinking rationally" approach to AI.
S2	The "AI winters", periods of reduced funding and interest, occurred in the 1950s, before the 1956 Dartmouth workshop. That workshop then revived the field and coined the term "artificial intelligence".
S3	A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance. Hence a taxi that picks a route later blocked by an unforeseeable landslide was not rational.
S4	PEAS description of an automated taxi – Performance measure: steering, accelerator, brake, horn. Environment: roads, other traffic, pedestrians, weather. Actuators: safe, fast, legal and comfortable trip. Sensors: cameras, GPS, speedometer.
S5	Chess played without a clock is a partially observable, stochastic, episodic, static and discrete task environment.
S6	A model-based reflex agent keeps an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can therefore act sensibly even when the current percept does not reveal the full state of the world.

(a) Identify which statements are correct and which are incorrect. For each incorrect statement, quote the erroneous phrase or claim (one statement contains more than one error).

Statement	Correct / Incorrect	Erroneous phrase / claim
S1		
S2		
S3		
S4		
S5		
S6		

(b) Root cause analysis: for every error, explain the underlying misconception, i.e. which concept has been misunderstood and why the statement goes wrong.

Error (statement / part)	Root cause (misconception)

Error (statement / part)	Root cause (misconception)

(c) Rewrite each faulty statement correctly and justify your correction. For a statement that is already correct, write "No change" and justify why.

Statement	Corrected statement	Justification
S1		
S2		
S3		
S4		
S5		
S6		

(d) Apply the corrected concepts to the following case. A self-driving taxi chooses a highway route that every available traffic report showed to be clear; an unforeseeable landslide then blocks the road.

(i) Using the four factors that decide whether an agent is rational, argue whether the taxi's choice was rational.

(ii) Name the properties of the taxi's task environment that make perfect foresight impossible.

(iii) Describe one situation in which a simple reflex taxi would behave poorly, and state what a model-based reflex taxi keeps in order to avoid it.

(i)	
(ii)	
(iii)	

Question 2: Problem-Solving Agents and Problem Formulation – Error Analysis of a Formulation

Topic: Problem-solving agents, problem formulation

A student formulates the **Missionaries and Cannibals** problem for a problem-solving agent. Three missionaries and three cannibals stand on the left bank of a river, with a boat that carries one or two people and cannot cross empty. The boat is on the left bank at the start. Missionaries must never be outnumbered by cannibals on either bank (a bank with no missionaries is safe). The objective is to move everyone to the right bank using the **fewest crossings**.

Component	Student's formulation
State representation	(M, C) = number of missionaries and cannibals on the left bank
Initial state	(3, 3)
Actions	Take the boat across with 1M, 2M, 1C, 2C or 1M + 1C
Goal test	M = C
Legality check	A state is legal if C ≤ M on the left bank (or M = 0 there)
Path cost	Number of people carried in the crossing
Note N1	The agent works in the order: search → goal formulation → problem formulation → execution.
Note N2	The state space contains 16 states, of which 10 are legal.

(a) For each component and note, state whether it is correct or incorrect; for each incorrect one, state what is wrong.

Component / Note	Correct / Incorrect	What is wrong
State representation		
Initial state		
Actions		
Goal test		
Legality check		
Path cost		
Note N1		
Note N2		

(b) Root cause analysis: for every error, explain the underlying cause, i.e. which principle of problem formulation or of the problem-solving agent has been violated.

Error (component / note)	Root cause

Error (component / note)	Root cause

(c) Give the corrected version of every faulty component or note and justify each correction.

Component / Note	Corrected formulation	Justification

(d) Apply your corrected formulation.

(i) Complete the table of boat loads available from the initial state, writing each resulting state as (M, C, boat side) and whether it is legal, with the reason.

Boat load	Resulting state (M, C, boat)	Legal? Reason
1M		
2M		
1C		
2C		
1M + 1C		

(ii) Give one state that the student's legality check accepts although it is actually illegal.

(iii) Using your corrected path cost, state the cost of an optimal solution.

(ii)	
(iii)	

Question 3: Searching for Solutions and Uninformed Search – Error Analysis of Search Traces

Topic: Searching for solutions, BFS, DFS, uniform-cost search

A student ran BFS, DFS and uniform-cost search (UCS) on the weighted undirected graph below. The start node is S and the goal node is G. **Conventions:** graph search with an explored set; neighbours expanded alphabetically; goal test when a node leaves the frontier; ties broken alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them. **BFS:** do not add a node that is already explored or already in the queue. **DFS:** use a stack with its top on the left; push children so that the alphabetically first child is popped first; do not push a node that is already explored or already on the stack. **UCS:** write entries as Node:g(parent) and replace an entry whenever a cheaper path to a frontier node is found.

Edge	Cost	Edge	Cost
S – C	3	B – G	3
S – D	3	D – E	2
A – B	2	D – G	6
A – C	6	E – G	3
B – D	4		

Student's BFS trace (queue initially [S])

Iteration	Node Dequeued	Queue after Expansion	Explored Set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[B, E, G, A]	{S, C, D}
4	B	[E, G, A]	{S, C, D, B}
5	E	[G, A]	{S, C, D, B, E}
6	G (goal)	[A]	{S, C, D, B, E}

BFS result: path S – D – G, cost 9, 6 nodes removed from the frontier.

Student's DFS trace (stack initially [S], top on the left)

Iteration	Node Popped	Stack after Pushing Children
1	S	[D, C]
2	D	[G, E, B, C]
3	G (goal)	[E, B, C]

DFS result: path S – D – G, cost 9, 3 nodes removed from the frontier.

Student's UCS trace

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), A:9(C), G:9(D)
5	B	7	A:9(C), G:9(D)
6	A	9	G:9(D)
7	G (goal)	9	–

UCS result: path S – D – G, cost 9, 7 nodes removed from the frontier.

Student's remarks

No.	Remark
R1	An explored set is not needed in BFS because BFS never revisits a node, so the Explored Set column is unnecessary.
R2	All three strategies returned S – D – G with cost 9, so all three are optimal and the traces are confirmed correct.
R3	DFS removed the fewest nodes from the frontier (3), so DFS is the best strategy for this graph.

(a) Examine the three traces and the remarks. List every error, giving the trace and iteration (or the remark number) and what is wrong. An error that is only carried forward counts once.

No.	Where (trace / iteration / remark)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the rule or concept that was violated and how it led to the wrong entry or conclusion.

No.	Root cause (rule or concept violated)
1	
2	
3	
4	
5	
6	
7	

(c) Write the corrected traces and justify each correction.

(i) BFS: the corrected queue after iteration 3 and the full order in which nodes are dequeued.

Queue after iteration 3	
Order of dequeued nodes	

(ii) DFS: the corrected trace.

Iteration	Node Popped	Stack after Pushing Children
1		
2		
3		
4		
5		

(iii) UCS: the corrected trace.

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1			
2			
3			
4			
5			
6			

Justification of the corrections	

(d) Apply the corrected traces.

(i) Complete the comparison table.

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
BFS				
DFS				
UCS				

(ii) Use iteration 3 of the corrected BFS trace to explain why an explored set is needed in graph search.

(ii)	

Question 4: Search with Partial Information – Error Analysis of Belief-State Computation

Topic: Problem formulation, search with partial information (sensorless and percept-based belief-state search)

A vacuum agent lives in a two-square world (Left square, Right square), each square being Clean (C) or Dirty (D). Actions: Left, Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suck cleans the current square). The goal is that both squares are Clean. Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

Task A – sensorless agent. The agent knows it is in the Left square but nothing about the dirt, so its initial belief state is {S1, S3, S5, S7}. A student applied the plan [Suck, Right, Suck]:

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S6, S8}

Task B – agent with a local sensor that reports [location, status of the current square]. Starting from the belief state of all eight states, the agent receives the percept [Left, Dirty], performs Suck, receives [Left, Clean], performs Right and receives [Right, Dirty]. The student's table:

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S2, S3, S6}
1	Suck	{S4, S5, S7, S8}	[Left, Clean]	{S5, S7}
2	Right	{S6, S7}	[Right, Dirty]	{S6}

Student's conclusions

No.	Conclusion
C1	Task A: the final belief state still contains two states, so the agent can never know where it is. No plan can guarantee the goal when the agent has no sensors; hence sensorless problems cannot be solved.
C2	Task B: the belief state is now the singleton {S6}, so the agent knows its location and the dirt status, and the goal has been reached.
C3	With 8 physical states there are only 8 belief states (one per physical state), so searching over belief states is no harder than searching over physical states.

(a) Examine the tables and the conclusions. List every error, giving the task and step (or the conclusion number) and what is wrong. An error that is only carried forward from an earlier wrong row counts once.

No.	Where (task / step / conclusion)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the concept or rule of belief-state search that was misapplied.

No.	Root cause (concept or rule misapplied)
1	
2	
3	
4	
5	
6	
7	

(c) Write the corrected tables and conclusions, with justification.

(i) Task A: the corrected belief states.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	
2	Right	
3	Suck	

(ii) Task B: the corrected belief states after each action and each percept.

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
1	Suck		[Left, Clean]	
2	Right		[Right, Dirty]	

(iii) Corrected conclusions C1 and C2, with justification.

C1	
C2	

(d) Apply the corrected concepts.

(i) How many different belief states are possible when there are 8 physical states? Why does this make belief-state search harder than search over physical states?

(i)	

(ii) For the initial belief state {S1, S3, S5, S7}, apply the plan [Right, Suck], write the belief state after each action, and state whether it is a solution.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Right	
2	Suck	
	Solution? Why?	

Rubrics for Error Analysis

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Error Identification	**20**	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	**30**	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	**25**	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	**15**	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	**10**	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Answer 1

(a) Identification of correct and incorrect statements

Statement	Correct / Incorrect	Erroneous phrase / claim
S1	Incorrect	(1) "judges a machine to be intelligent if its internal reasoning always follows the laws of logic" (2) "the standard example of the 'thinking rationally' approach" (follows from error 1; counted once)
S2	Incorrect	(1) "AI winters ... occurred in the 1950s, before the 1956 Dartmouth workshop" (2) "That workshop then revived the field" (the part "coined the term artificial intelligence" is correct)
S3	Incorrect	(1) "always achieves the best possible outcome because it knows the actual result of every action in advance" (2) "was not rational" (conclusion carried forward from error 1)
S4	Incorrect	Two errors (components swapped): "Performance measure: steering, accelerator, brake, horn" and "Actuators: safe, fast, legal and comfortable trip". Environment and Sensors are correct.
S5	Incorrect	Three errors: "partially observable", "stochastic", "episodic". ("static" and "discrete" are correct.)
S6	Correct	None

(b) Root cause analysis

Error (statement / part)	Root cause (misconception)
S1	The Turing Test is a behavioural test: it checks whether a human interrogator can tell the machine from a human in conversation. This is the "acting humanly" approach. The student confused it with "thinking rationally" (the laws-of-thought / logic approach), which concerns correct reasoning, not observable behaviour.
S2 (1)	Chronology error. The 1956 Dartmouth workshop is the birth of AI, so nothing called an AI winter can precede it. The AI winters came later (mid-1970s, and late 1980s to early 1990s) after over-optimistic promises failed.
S2 (2)	Dartmouth founded and named the field; it did not revive it. Revival belongs to later events (expert systems in the 1980s, machine learning and neural networks from the late 1980s onward).
S3 (1)	Rationality was confused with omniscience. A rational agent maximises expected performance given its percept sequence and built-in knowledge; it cannot know actual outcomes in advance. Rationality is not perfection.
S3 (2)	Wrong conclusion drawn from the wrong definition. Rationality is judged by the decision made with available information, not by the outcome of an unforeseeable event.
S4	PEAS roles were mixed up: Performance measure = the criteria for success (safe, fast, legal, comfortable); Actuators = the means of acting (steering, accelerator, brake, horn).
S5 (partially observable)	Chess shows the complete board to both players at all times, so it is fully observable. Hidden information (as in poker) was confused with a large search space.
S5 (stochastic)	No chance element exists in chess: the next state is fully determined by the current state and the move, so it is deterministic. Difficulty of prediction was confused with randomness.

Error (statement / part)	Root cause (misconception)
S5 (episodic)	Each move affects all later positions, so the task is sequential. Episodic means every percept-action episode is independent of the previous ones.
S6	No error. (Correct description of a model-based reflex agent.)

(c) Corrected statements and justification

Statement	Corrected statement	Justification
S1	The Turing Test (1950) judges a machine to be intelligent if a human interrogator cannot reliably distinguish its conversational responses from those of a human. It is the standard example of the "acting humanly" approach.	The test depends only on external behaviour, not on the quality of internal reasoning.
S2	The 1956 Dartmouth workshop coined the term "artificial intelligence" and launched the field. The "AI winters", periods of reduced funding and interest, came afterwards (mid-1970s and late 1980s to early 1990s).	Winters followed unmet expectations; they cannot precede the founding of the field.
S3	A rational agent selects the action expected to maximise its performance measure, given its percept sequence and prior knowledge. It need not know actual outcomes (it is not omniscient). A taxi whose route is later blocked by an unforeseeable landslide was still rational.	Rationality concerns expected performance with available information, not the actual outcome.
S4	Performance measure: safe, fast, legal and comfortable trip. Environment: roads, other traffic, pedestrians, weather. Actuators: steering, accelerator, brake, horn. Sensors: cameras, GPS, speedometer.	Performance measure states what is desired; actuators are the means of acting.
S5	Chess played without a clock is a fully observable, deterministic, sequential, static and discrete (multiagent, competitive) task environment.	Whole board is visible; moves have no random effects; each move affects later positions; no clock means the world does not change while the agent deliberates; finite discrete states and moves.
S6	No change	A model-based reflex agent maintains an internal state using a transition model (how the world evolves, effects of its actions), so it can handle partial observability.

(d) Application to the self-driving taxi

(i) Rationality: The four factors are (1) the performance measure (safe, fast trip), (2) the agent's prior knowledge of the environment, (3) the actions it can perform, and (4) its percept sequence to date. Every available traffic report showed the highway as clear, so given its percepts, choosing that route maximised the expected performance. The landslide was unforeseeable and so could not be part of the percept sequence or prior knowledge. Hence the choice was rational. The bad outcome shows that the agent is not omniscient, not that it was irrational.

(ii) Properties: Partially observable (it cannot sense the stability of hillsides or future events), stochastic (outcomes of actions are uncertain), dynamic (the road changes while the agent acts), sequential, continuous and multiagent. Partial observability, stochasticity and dynamism are what make perfect foresight impossible.

(iii) Reflex vs model-based: A simple reflex taxi acts only on the current percept. For example, when a car in the next lane is in its blind spot, the current percept shows nothing, so the reflex taxi changes lanes and collides. A model-based reflex taxi keeps an internal state: its previous observations (a car seen a moment

ago in the mirror), a model of how the world evolves (the other car keeps moving at about the same speed), and a model of the effect of its own actions (turning the wheel moves the taxi sideways). This lets it infer the hidden car and wait.

Answer 2

(a) Correct / incorrect classification

Component / Note	Correct / Incorrect	What is wrong
State representation	Incorrect	(M, C) omits the boat position. The legal moves depend on which bank the boat is on, so two different situations collapse into one state.
Initial state	Incorrect	(3, 3) is incomplete (boat side missing). Carried forward from the state representation; it should be (3, 3, L).
Actions	Correct	The five boat loads (1M, 2M, 1C, 2C, 1M+1C) cover every non-empty load of at most two people. (Applicability and direction depend on the boat side.)
Goal test	Incorrect	M = C is true at the initial state (3, 3) and at many mid-way states such as (2, 2). The goal is that everyone is on the right bank.
Legality check	Incorrect	Checks only the left bank. The right bank must also satisfy "no missionaries, or cannibals do not outnumber missionaries".
Path cost	Incorrect	The objective is to minimise the number of crossings, so each crossing should cost 1, not the number of people carried.
Note N1	Incorrect	Order is wrong. The correct order is goal formulation, problem formulation, search, execution.
Note N2	Incorrect	16 states and 10 legal hold only for the incomplete (M, C) representation. With the boat side, there are 4 x 4 x 2 = 32 states, of which 20 are legal.

(b) Root cause analysis

Error (component / note)	Root cause
State representation	A state must contain all information needed to decide the legal actions and their results. Leaving out the boat position violated this principle (abstraction removed a relevant detail).
Initial state	Follows from the incomplete state description (carried forward, counted with the state representation).
Goal test	A goal test must be true only for the goal states. The student wrote a property that is also true at non-goal states, instead of specifying the goal state (0, 0, R) as stated in the objective.
Legality check	The safety constraint applies to both banks. Only one bank was checked, so states violating the constraint on the right bank are accepted.
Path cost	The path cost function must reflect the stated objective (fewest crossings). Using people carried measures a different quantity, so the cheapest solution would not be the one with the fewest crossings.
Note N1	A problem-solving agent must first formulate the goal (which states are acceptable), then formulate the problem (states and actions), and only then search. Search cannot begin without a problem; execution comes last.
Note N2	State-space size was counted from the flawed representation. Each (M, C) pair exists with the boat on either bank, so counts must be doubled.

(c) Corrected formulation and justification

Component / Note	Corrected formulation	Justification
State representation	(M, C, B): M, C = missionaries and cannibals on the left bank; B in {L, R} = boat side.	Contains everything needed to compute legal actions.
Initial state	(3, 3, L)	Boat starts on the left bank.
Actions	Load (m, c) in {(1,0), (2,0), (0,1), (0,2), (1,1)}, applicable only if the bank with the boat has at least m missionaries and c cannibals. If B = L: (M - m, C - c, R); if B = R: (M + m, C + c, L).	Boat carries 1 or 2 people and cannot cross empty; the load cannot exceed the people on the boat bank.
Goal test	State = (0, 0, R)	All six people (and the boat) on the right bank.
Legality check	Left: M = 0 or C <= M. Right: (3 - M) = 0 or (3 - C) <= (3 - M). Both must hold.	Constraint applies to both banks.
Path cost	Each crossing costs 1; path cost = number of crossings.	Matches the objective of fewest crossings.
Note N1	Goal formulation, then problem formulation, then search, then execution.	Standard problem-solving agent cycle.
Note N2	32 states in total, 20 of them legal.	10 legal (M, C) pairs (M = 0 or 3 with any C, plus (1,1) and (2,2)), times 2 boat positions.

(d) Application of the corrected formulation

(i) Boat loads from the initial state (3, 3, L)

Boat load	Resulting state (M, C, boat)	Legal?	Reason
1M	(2, 3, R)	No	Left bank: 2 missionaries, 3 cannibals; cannibals outnumber missionaries.
2M	(1, 3, R)	No	Left bank: 1 missionary, 3 cannibals; cannibals outnumber missionaries.
1C	(3, 2, R)	Yes	Left 3M, 2C is safe; right bank 0M, 1C (no missionaries, so safe).
2C	(3, 1, R)	Yes	Left 3M, 1C is safe; right bank 0M, 2C (safe).
1M + 1C	(2, 2, R)	Yes	Left 2M, 2C is safe; right bank 1M, 1C is safe.

(ii) State wrongly accepted by the student's check: (M, C) = (2, 1), e.g. (2, 1, R). The left bank (2M, 1C) satisfies C <= M, so the student accepts it, but the right bank holds 1M and 2C, where cannibals outnumber missionaries, so it is illegal. (1, 0) and (2, 0) are further examples.

(iii) Optimal cost: The optimal solution needs 11 crossings, so the path cost is 11 (cost 1 per crossing). One optimal sequence: (3,3,L) to (3,1,R) to (3,2,L) to (3,0,R) to (3,1,L) to (1,1,R) to (2,2,L) to (0,2,R) to (0,3,L) to (0,1,R) to (0,2,L) to (0,0,R).

Answer 3

(a) Errors in the traces and remarks

No.	Where (trace / iteration / remark)	What is wrong
1	BFS, iteration 3	Queue after expanding D is written [B, E, G, A]. BFS is FIFO, and A was added earlier (iteration 2), so the queue must be [A, B, E, G]. (Later BFS rows, i.e. dequeuing B instead of A in iteration 4, the explored sets, "6 nodes removed" and A left unexplored in the queue, are carried forward.)
2	DFS, iteration 1	Stack after pushing the children of S is written [D, C]. The alphabetically first child (C) must be on top, so the stack is [C, D]. (The wrong path S-D-G, cost 9 and 3 nodes removed are carried forward.)
3	UCS, iteration 4	When E (g = 5) is expanded, G is reached with g = 5 + 3 = 8, which is cheaper than G:9(D) in the queue. The entry should be replaced by G:8(E); the student left G:9(D). (A expanded in iteration 6, goal at 9, path S-D-G and 7 nodes removed are carried forward.)
4	Remark R1	Claims BFS does not need an explored set. In graph search it does, since a node that has already left the queue (e.g. S, reached again from D) would otherwise be re-added. The student also contradicted the claim by using an Explored Set column.
5	Remark R2 (part 1)	"All three are optimal" is wrong. Cost 9 is not minimal: S-D-E-G costs 8. Only UCS is guaranteed optimal for varying step costs.
6	Remark R2 (part 2)	"Agreement of the three results confirms the traces are correct" is invalid. The traces contain errors, and identical final answers do not validate the intermediate steps.
7	Remark R3	"DFS removed fewest nodes so it is best" is wrong. Node count is not the criterion for the best strategy; DFS actually removes 5 nodes (not 3) and returns the worst path (cost 14).

(b) Root cause analysis

No.	Root cause (rule or concept violated)
1	BFS uses a FIFO queue: nodes enter at the back and leave from the front, so earlier-discovered nodes (A) are dequeued before later ones (B, E, G). The student put A at the back after the new children, which broke FIFO order. This changed every later dequeue.
2	DFS uses a stack (LIFO) with the alphabetically first child to be popped first. Pushing C then D with the top on the left gives [C, D]; the student reversed the order. This sent the search down the D branch and changed the path, cost and node count.
3	UCS must update a frontier entry whenever a cheaper path is found (Node:g(parent) replacement rule). The cheaper path to G via E (cost 8) was missed. Because of the stale cost 9, UCS expanded A and returned a non-optimal result.
4	Tree search vs graph search confusion. The frontier check only prevents duplicates of nodes currently waiting; the explored set is the only protection against re-adding nodes already expanded. Undirected edges and cycles make revisits happen constantly.
5	Equating "same answer" with "optimal". BFS minimises the number of steps, not cost; DFS has no optimality guarantee; only UCS is optimal because it expands in order of g(n). Student failed to compare against the true optimum.
6	Invalid verification: matching outputs prove nothing about correctness. A trace is verified by checking each step against the rules (queue discipline, explored set, cost updates).

No.	Root cause (rule or concept violated)
7	Using an irrelevant criterion. Strategies are judged by completeness, optimality, and time and space cost. Fewer removals on one graph does not make DFS "best", especially when the path it returns is the most expensive.

(c) Corrected traces

(i) BFS

Iteration	Node dequeued	Queue after expansion	Explored set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[A, B, E, G]	{S, C, D}
4	A	[B, E, G]	{S, C, D, A}
5	B	[E, G]	{S, C, D, A, B}
6	E	[G]	{S, C, D, A, B, E}
7	G (goal)	[ ]	{S, C, D, A, B, E}

Item	Answer
Queue after iteration 3	[A, B, E, G]
Order of dequeued nodes	S, C, D, A, B, E, G

(ii) DFS

Iteration	Node popped	Stack after pushing children
1	S	[C, D]
2	C	[A, D] (S explored; A new)
3	A	[B, D] (C explored; B new)
4	B	[G, D] (A explored; D already on stack, not pushed; G new)
5	G (goal)	[D]

(iii) UCS

Iteration	Node expanded	g(n)	Priority queue after expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), G:8(E), A:9(C) (G replaced: 8 < 9)
5	B	7	G:8(E), A:9(C) (via B: A = 9 not cheaper, G = 10 not cheaper)
6	G (goal)	8	A:9(C)

Justification of the corrections: BFS follows FIFO order, so A (added in iteration 2) is dequeued before B, E, G. DFS pushes C above D so C is popped first; D is not pushed again from B because it is already on the

stack. UCS replaces G:9(D) by G:8(E) since g(G) = g(E) + c(E, G) = 5 + 3 = 8. The goal G is removed with g = 8, giving S-D-E-G; A is never expanded. In the UCS rows, ties are broken alphabetically (C before D; A before G).

(d) Application of the corrected traces

(i) Comparison table

Strategy	Path found	Path cost	Nodes removed from frontier	Optimal here?
BFS	S - D - G	3 + 6 = 9	7	No (optimal cost is 8)
DFS	S - C - A - B - G	3 + 6 + 2 + 3 = 14	5	No
UCS	S - D - E - G	3 + 2 + 3 = 8	6	Yes

(ii) Need for an explored set: In iteration 3, D is dequeued and its neighbours are B, E, G and S. B, E, G are new, but S was already expanded in iteration 1; it is no longer in the queue, so a queue-membership check alone would not catch it and S would be added again. S would then be expanded again, producing C and D again, and so on, giving repeated work (and an infinite loop on cyclic graphs). The explored set {S, C, D} records already-expanded nodes so that S is not re-added, which is why graph search needs it.

Answer 4

(a) Errors in the tables and conclusions

No.	Where (task / step / conclusion)	What is wrong
1	Task A, step 3 (Suck)	Belief written as {S6, S8}. Suck in S6 = (R, C, D) cleans the right square and gives S8; S8 stays S8. The belief is {S8}.
2	Task B, step 0, percept [Left, Dirty]	Updated belief {S1, S2, S3, S6} contains S2 and S6 (agent in the Right square). Only states with agent in Left and Left square dirty remain: {S1, S3}. (Predicted set {S4, S5, S7, S8} in step 1 is carried forward.)
3	Task B, step 2 (Right), predicted	Predicted belief {S6, S7} is wrong: Right moves S7 = (L, C, C) to S8 = (R, C, C), not S7. Correct predicted belief is {S6, S8}.
4	Conclusion C1 (part 1)	"The agent can never know where it is" is false. The belief state reduces to {S8}, a single state, and the agent knew its location (Left, then Right) throughout. (The belief of two states was caused by error 1.)
5	Conclusion C1 (part 2)	"No plan can guarantee the goal; sensorless problems cannot be solved" is false. [Suck, Right, Suck] reaches {S8}, so it is a guaranteed solution (coercion).
6	Conclusion C2	Belief {S6} means the agent knows the state, but the goal is NOT reached: S6 = (R, C, D) has a dirty Right square. The agent still needs to Suck.
7	Conclusion C3	8 physical states give 2^8 = 256 belief states (any subset), not 8. Belief-state search is therefore much harder than physical-state search.

(b) Root cause analysis

No.	Root cause (concept or rule misapplied)
1	Applying the action to a belief state means applying it to every state in the set. The student did not apply Suck to S6, forgetting that the Right square is still dirty in S6 and Suck cleans the current square.

No.	Root cause (concept or rule misapplied)
2	The percept update step must filter the belief state to the states consistent with the percept. [Left, Dirty] is consistent only with agent location L and a dirty Left square. The student kept states with the agent in R and S6, where the Left square is clean.
3	The transition model was misread: Right moves the agent from the Left to the Right square; it is a no-op only in the Right square. The student treated S7 as unaffected.
4	Misunderstanding of belief states. Belief shrinks as actions are applied (coercion), and knowing the location is separate from knowing the dirt; here every state in the initial belief has the agent in L.
5	Sensorless problems are solvable by searching in belief-state space: a plan is a solution if every state in the final belief is a goal state. The student wrongly required observations.
6	Confusing "knowing the state" (belief is a singleton) with "being in a goal state". The goal test is that every state in the belief is a goal state (both squares clean).
7	A belief state is a subset of physical states, so n states give 2^n belief states. The student counted one belief state per physical state (only singleton beliefs).

(c) Corrected tables and conclusions

(i) Task A

Step	Action	Belief state after action
0 (start)		{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S8}

(ii) Task B

Step	Action	After action (predicted)	Percept	After percept (updated)
0 (start)		{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S3}
1	Suck	{S5, S7}	[Left, Clean]	{S5, S7}
2	Right	{S6, S8}	[Right, Dirty]	{S6}

(iii) Corrected conclusions

C1: Task A: the final belief state is {S8}, a single state in which both squares are clean. Sensorless problems can be solved: the plan [Suck, Right, Suck] reaches the goal from every state in the initial belief {S1, S3, S5, S7}, so it is a solution. The agent also knows where it is (Left at the start, Right after the move).

C2: Task B: the belief {S6} is a singleton, so the agent knows its state exactly (Right square, Left clean, Right dirty), but the goal has not been reached because the Right square is dirty. The agent must perform Suck, which gives {S8}.

(d) Application of the corrected concepts

(i) Number of belief states: Each belief state is a subset of the 8 physical states, so there are 2^8 = 256 possible belief states (255 non-empty). Search over belief states is harder because the space is exponentially larger (2^n vs n), each node is a set of states, and every action and percept update must be applied to every state in the set.

(ii) Plan [Right, Suck] from {S1, S3, S5, S7}

Step	Action	Belief state after action
0 (start)		{S1, S3, S5, S7}
1	Right	{S2, S4, S6, S8}
2	Suck	{S4, S8}

Solution? No. The final belief {S4, S8} contains S4 = (R, D, C), where the Left square is still dirty. A plan is a solution only if every state in the final belief is a goal state, and S4 is not.